Signal Review Guide

Beginner-friendly rules for aggregate safety review.

1. Start with data quality

• Check duplicate indicators if available.

• Check missing product/event/outcome fields.

• Confirm event coding consistency.

• Confirm the observation window and denominator.

2. Descriptive review

• Count total reports by product.

• Summarize serious cases and outcomes.

• Review event frequencies and temporal/reporting patterns.

• Separate case counts from incidence; spontaneous report counts do not provide exposure-based incidence.

3. Potential signal

• Identify a product-event combination that appears notable.

• Calculate a transparent descriptive disproportionality measure when the dataset supports it.

• Treat the result as a prioritization signal, not proof of causation.

4. Alternative explanations

• Confounding by indication or patient characteristics.

• Differential reporting or stimulated reporting.

• Changes in exposure, prescribing, coding, or reporting practices.

• Duplicate or incomplete cases.

• Background disease or concomitant medicines.

5. Follow-up

• Review individual cases and chronology.

• Verify duplicates and coding.

• Assess dechallenge/rechallenge where available.

• Check clinical/laboratory evidence.

• Compare with external safety information and other data sources.

• Consider whether risk-minimization messaging or monitoring needs evaluation.

Official learning link

EMA GVP Module IX and EudraVigilance guidance are suitable references for learning about signal detection and evaluation.